

TSMC: The Observable Engine and the Unidentified Moat

Taiwan Semiconductor Manufacturing Company — TWSE: 2330 · NYSE: TSM (ADR)

Classification	Investment research I Semiconductor / Foundry
Designation	No rating — evidence-based research
Valuation date	11 August 2026
Research review through	15 August 2026; operating disclosures through Q2 2026
ADR price	US\$422
Market capitalisation	~US\$2.19tn
Conclusion	Public evidence does not support an actionable edge at the current price

This report issues no rating and no fair-value estimate. Its conclusion concerns the sufficiency of available evidence, not a judgment that the shares are cheap or expensive. See Section 1.

15 August 2026 is the research review and publication cutoff, not a new financial dataset. No primary disclosure was issued between the 11 August 2026 valuation date and this cutoff; all figures, tables and conclusions are unchanged as a result.

1. Executive Investment View

The available public evidence cannot establish an actionable edge in TSMC at US\$422 per ADR. This is a conclusion about the evidence rather than about the company. TSMC is not judged overvalued, undervalued or fairly valued here; the analysis simply does not reach a threshold at which an informed position could be taken on the basis of information the market does not already hold.

The reason is specific. TSMC's economic engine is unusually observable, and much of it reduces to arithmetic. Customer aggregation converts into utilisation, utilisation into fixed-cost absorption, absorption into surplus, and surplus into reinvestment under a capital rule that has now been tested through a full cycle. Two of those links are accounting identities rather than competitive advantages, and this report is careful not to let them become one. What sits underneath — the manufacturing capability that customers actually pay for — cannot be identified from public evidence. No foundry discloses yield, defect density or cost per good die at any node, and there is no failed TSMC node against which a mechanism could be tested.

That leaves the investment question resting on the valuation, and the valuation rests on a discount rate this framework has no defensible basis to set. At an 8% discount rate with 4% terminal growth, management's stated 25% revenue objective clears the current price with room. At 9% and 3%, the same model requires a revenue CAGR near 35% through 2030 — roughly US\$551bn of revenue against US\$122bn in FY2025. Terminal value accounts for 86–89% of the discounted cash flow output in every case run, which is a structural consequence of the narrow spread between discount rate and terminal growth rather than a modelling choice.

The most instructive number in the 2026 data is a decomposition that inverts depending on the comparison. Measured year on year, H1 2026 revenue growth of 35.6% was 65% volume: shipments rose 22.0% while blended revenue per wafer rose 11.2%. Measured sequentially, from Q1 to Q2, the split reverses to 34% volume: shipments rose 3.9% and revenue per wafer 7.8%. Both are correct. The year-on-year figure reflects capacity commissioned during 2025; the sequential figure reflects the N2 and N3 migration happening now. A thesis citing only the first is describing last year's capacity build rather than the current engine.

What follows sets out where the observable economics end, why the technical moat remains unidentified, what today's price requires, and what evidence would change the decision.

A note on evidence. Claims below are graded, though not tagged inline: **A** disclosed or directly observed, **B** calculated from A-grade inputs, **C** strong inference, **D** plausible but unverified, **E** unknown from public evidence, **Guided** a management forward statement, **Assumed** an explicit modelling choice. No grade is upgraded for narrative convenience, and where a conclusion rests on inference or assumption the text says so at the point it matters. Full definitions are in Appendix G.

WHAT THIS REPORT DOES NOT CLAIM

This report does not claim TSMC is overvalued. It does not claim TSMC is undervalued. It does not identify the technical mechanism behind TSMC's process capability — that mechanism is graded unidentified from public evidence throughout. It does not assign a probability to geopolitical disruption of Taiwan-based operations. It does not treat utilisation-driven margin expansion as a moat; utilisation and fixed-cost absorption are accounting identities, not competitive advantages. And it does not treat the discounted cash flow output as an estimate of intrinsic value — it is used here as a requirement engine, showing what must be true of growth and discount rate, not what the business is worth.

2. What the Market Is Paying For

At roughly US\$2.19tn of market capitalisation, the price can be translated into operating requirements. Holding the base-case cost and capital structure constant and varying only the revenue growth rate, at a 9% discount rate with 3% terminal growth:

Revenue CAGR to 2030	2030 revenue	Value vs. market
20%	US\$342bn	−33%
25% (stated objective)	US\$404bn	−24%
30%	US\$473bn	−13%
35%	US\$551bn	−1%
40%	US\$638bn	+12%

Calculated from the base-case model; see Appendix B.

At 9% and 3%, today's price requires a revenue CAGR of approximately 35% through 2030. That is ten points above management's own long-term objective and above any four-year period in the company's history. Whether this is demanding or reasonable depends entirely on the discount rate applied, which is the point.

Running the same question across scenarios and discount assumptions:

Case	10% / 3%	9% / 3%	8% / 4%
Bear	−63%	−58%	−41%
Base, 20% CAGR	−41%	−32%	−3%
Base, 25% CAGR	−34%	−24%	+11%
Bull, 30% CAGR	−29%	−18%	+19%

Only the 8%/4% column clears, and only if growth meets or exceeds the stated objective. At 20% growth, even 8%/4% leaves the shares roughly at fair value rather than cheap.

The inputs that dominate this result are not the operating ones. Revenue, margins, shipments and capital expenditure for FY2026 are observed or guided and carry limited weight in a five-year model. The 2027–30 revenue path, the terminal growth rate and the discount rate are all assumptions, and the last two dominate. A useful discipline is to treat the discounted cash flow as a requirement engine rather than a valuation: it identifies what must be true, not what the business is worth.

Two thresholds follow. The price would look plainly cheap on sustained revenue growth above 30% through 2028 combined with a defensible sub-9% discount rate, most plausibly justified by a leading-edge node entering volume production outside Taiwan — the only observable development that would reduce the concentration premium. It would look plainly expensive if growth settles at or below 20% while capital intensity holds in the mid-30s, which at 9%/3% is a 33% shortfall.

3. The Observable Economic Engine

Roughly 42% of TSMC's cost of revenue is depreciation. Unit economics are therefore governed less by what is spent running a fab than by how many saleable die emerge from an asset base already paid for. When output rises against a largely fixed asset base, cost per wafer falls mechanically.

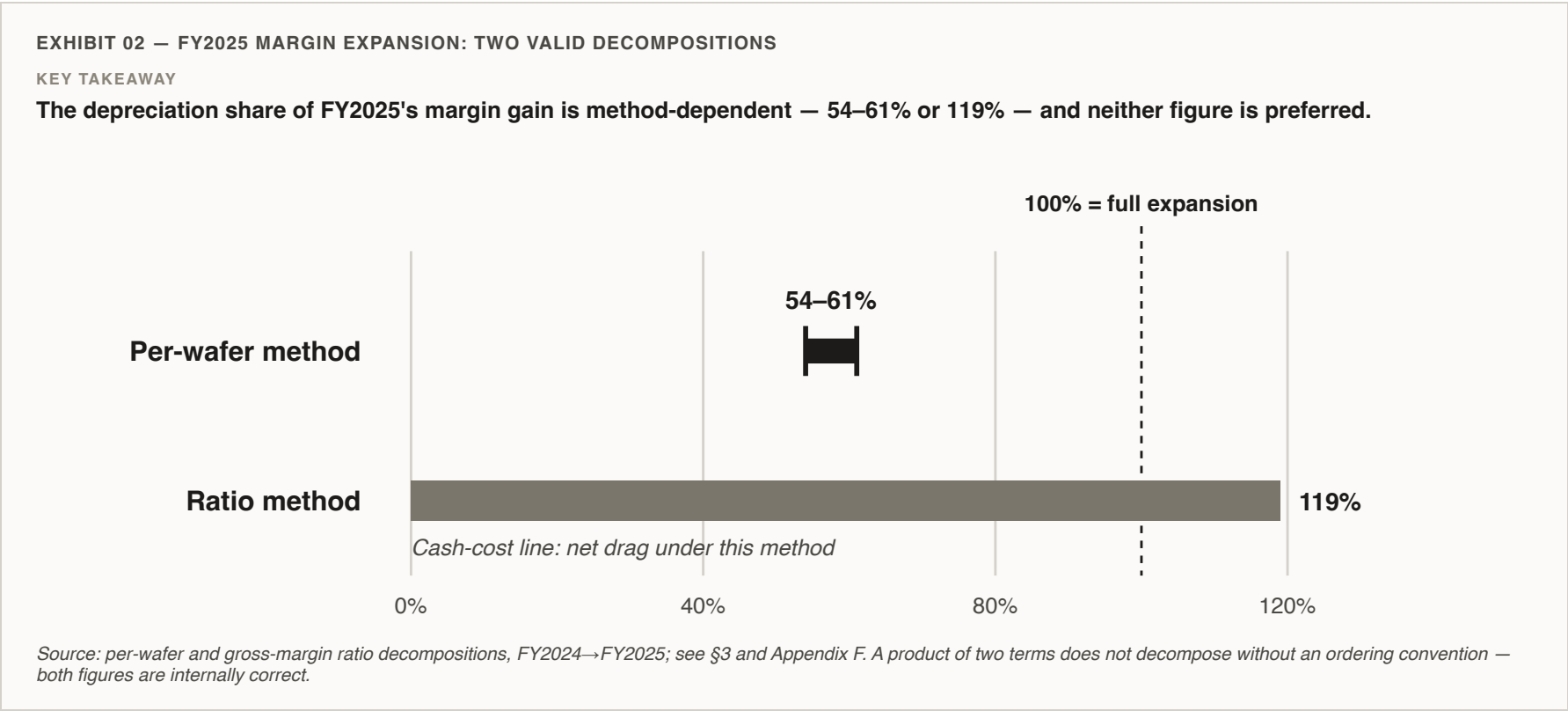
That is what happened in FY2025. Wafer shipments rose 16.4% against capacity that was approximately flat, and depreciation per wafer fell 10.8%.

NT\$ per 12-inch-equivalent wafer	FY2024	FY2025	Change
Cost of revenue	98,373	101,704	+3.4%
Depreciation in cost of revenue	47,645	42,511	−10.8%
Cash cost	50,728	59,193	+16.7%

Calculated from disclosed revenue, gross margin, wafer shipments and total depreciation, with depreciation allocated 92.8% to cost of revenue. The allocation derives from TSMC's functional-expense disclosure, in which the operating-expense share of total depreciation and amortisation was 9.50%, 8.30%, 7.24% and 7.21% across FY2019–22; the FY2023–25 share is estimated at 7.2% and is an assumption.

This chain — aggregation to utilisation to fixed-cost absorption to surplus to reinvestment — is where TSMC's observable advantage lives. But the middle two links are arithmetic, not competitive advantages. High utilisation is a consequence of demand aggregation; falling depreciation per wafer is a consequence of high utilisation. Calling either a moat confuses an outcome with a cause, and the distinction matters because arithmetic reverses when the inputs reverse.

The share of FY2025's margin expansion attributable to depreciation is method-dependent, and no single figure is assigned here. A per-wafer decomposition holding depreciation per wafer constant gives 54–61%. A ratio decomposition, splitting gross margin into its two cost ratios, gives 119% — under which the cash-cost line was a net drag. Both are internally correct. A product of two terms does not decompose without an arbitrary ordering choice. What survives is that depreciation absorption accounted for the majority of the expansion and the cash-cost line was neutral to negative.



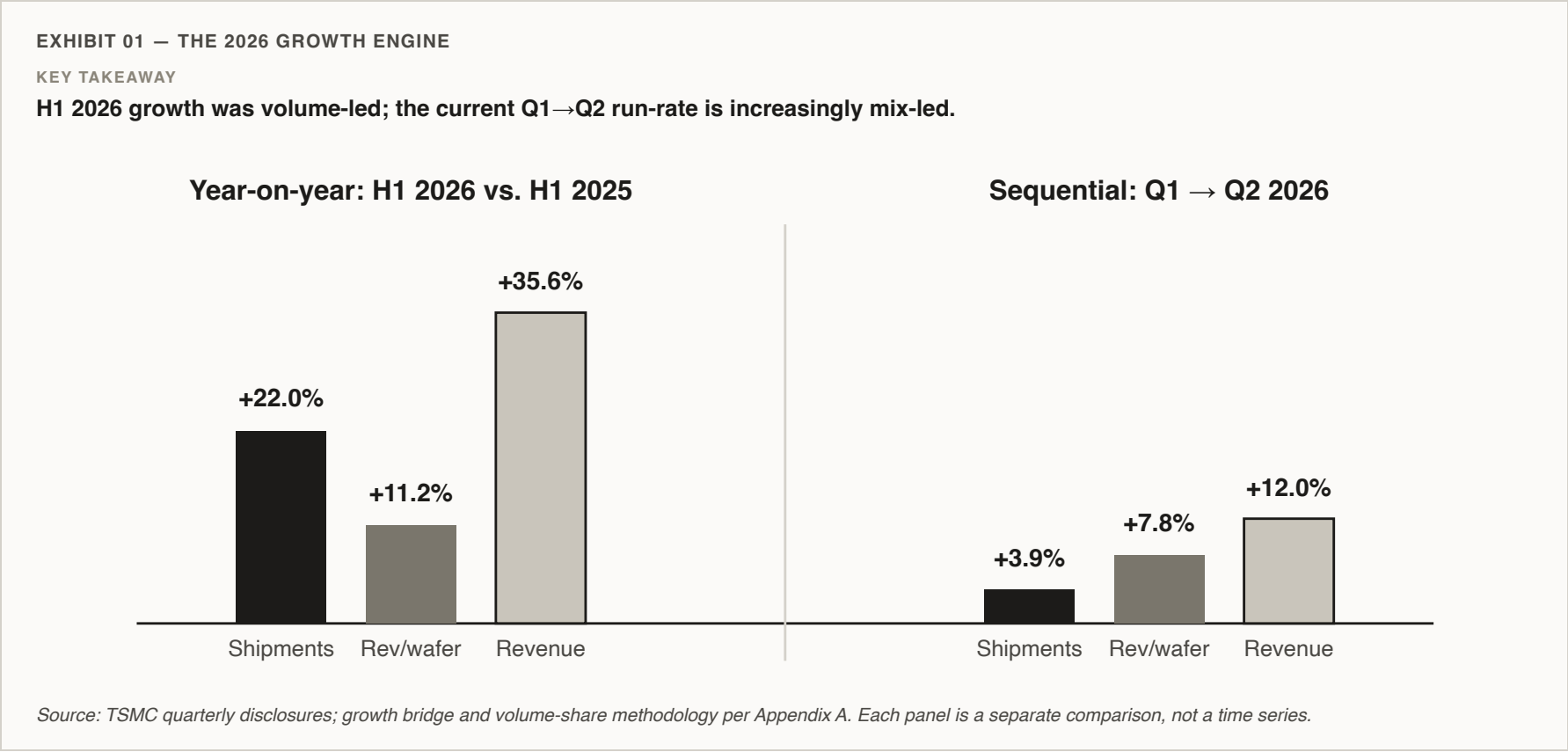
Price is not identified in the accounts. Converting node revenue shares into volume shares requires a relative node price vector that TSMC does not disclose. Across three plausible vectors the implied mix contribution ranges from +2.2% to +22.9%, leaving a price residual spanning −5% to +14%. That interval includes zero and it includes the entire observed increase in revenue per wafer. Corroborating this, price is absent from four consecutive quarterly gross-margin attributions, and the chief financial officer has stated that pricing benefits were "just enough to cover the inflation cost from tools, equipment, materials, labor."

4. Revenue, Wafer and Utilisation Economics

TSMC began disclosing quarterly wafer shipments in its earnings presentation, which allows the revenue bridge to be decomposed directly rather than inferred.

Comparison	Shipments	Revenue/wafer	Revenue	Volume share
H1 2026 vs. H1 2025	+22.0%	+11.2%	+35.6%	65%
Q2 2026 vs. Q1 2026	+3.9%	+7.8%	+12.0%	34%

The bridges reconcile: $1.220 \times 1.112 = 1.356$ against reported growth of 35.6%, and $1.039 \times 1.078 = 1.120$ against reported growth of 12.0%. Volume share is the log-ratio contribution; see Appendix A.



The two readings are not in conflict. The year-on-year comparison captures capacity that came into service across 2025 and is therefore a statement about last year's investment. The sequential comparison captures what is happening inside 2026: N2 moving from zero to 3% of wafer revenue and N3 reaching 30%. An analyst who cites only the year-on-year decomposition will conclude that growth is volume-led; one who cites only the sequential decomposition will conclude the opposite. Both figures belong in the record.

The sequential run-rate also settles a question that earlier drafts of this work treated as unresolved. Carrying the Q1-to-Q2 rates forward two quarters produces an implied Q3 of NT\$1,423bn against a guidance midpoint of NT\$1,437bn — a 1% gap — and a full-year figure of NT\$5,423bn, or roughly US\$171bn, consistent with guidance of slightly above 40% growth.

FY2026 implied from Q2 run-rate	Value
Revenue	NT\$5,423bn
Revenue, USD	~US\$171bn
Wafer shipments	17,693 Kpcs
Revenue per wafer	NT\$306,000

FY2026 guidance therefore requires no step-change in the business. It requires the current sequential pattern to persist for two more quarters, which is a considerably more testable proposition than a full-year growth rate.

Capacity is the weaker part of the picture. TSMC discloses annual capacity rounded to the nearest million 12-inch equivalents: approximately 16 million in FY2023, approximately 17 million in FY2024, and above 17 million in FY2025. FY2026 has not been disclosed. Implied utilisation therefore carries a band rather than a point: 74–78% in FY2024, 84–88% in FY2025, and 90–100% for H1 2026 annualised, depending on where FY2025 capacity sat within its rounding.

What the band does establish is that TSMC is operating at or near its practical ceiling, and management has been explicit that the next major tranche of capacity is "2028-2029 supply." The utilisation lever that produced FY2025's margin expansion is largely spent.

5. Margin and Depreciation Economics

FY2026 depreciation is guided at approximately high-teens percentage growth on FY2025's NT\$688.1obn, implying roughly NT\$812bn, or about 15.0% of revenue.

The forward schedule used in this report is a calibrated fit rather than a validated forecast, and the distinction is material. The model takes the form of a machinery share applied to a rolling five-year capital-expenditure average plus a building component. Testing it across 420 parameter combinations, twenty achieve better than 6% mean absolute error against the same historical observations — with four free parameters and five data points, a good fit is expected rather than informative. More importantly, a genuine out-of-sample test, calibrating on 2021, 2022 and 2024 and then predicting 2025 blind, errs by 8.4%. An error of that magnitude moves 2030 gross margin by roughly 1.4 points and equity value by roughly 4%.

One decomposition cannot be performed at all. TSMC's quarterly earnings materials disclose revenue, margins, operating expenses, the balance sheet and cash flows, but not depreciation and amortisation. Depending on how depreciation is phased within 2026, H1's 8.30-point gross-margin expansion splits as follows:

H1 share of FY depreciation	Depreciation	Cash cost
45% (back-loaded)	+4.91pp	+3.39pp
50% (even)	+3.34pp	+4.96pp
52.8% (FY2025 pattern)	+2.46pp	+5.84pp

A property, plant and equipment roll-forward does not resolve it. Net PP&E of NT\$3,954.68bn plus capital expenditure of NT\$496.00bn less NT\$4,302.88bn leaves a residual of NT\$147.80bn, but capital expenditure includes construction in progress that does not yet depreciate, the residual excludes right-of-use and intangible amortisation, and foreign-currency translation on overseas assets is not isolated.

This variable is unknown and remains so. Its decision-materiality is low: the FY2026 gross margin is observed rather than derived, and the phasing question affects only whether the expansion reverses on a depreciation schedule or on growth deceleration. That is worth roughly 3% of equity value through the terminal margin assumption.

6. Quality of Earnings

The 2Q26 income statement contains two features worth isolating.

First, the operating-income bridge does not close on gross profit less operating expenses. Gross profit of NT\$860.31bn less operating expenses of NT\$98.98bn gives NT\$761.33bn against reported income from operations of NT\$766.60bn. The reconciling line is other operating income and expenses, at NT\$5.27bn in 2Q26 and NT\$1.68bn in 1Q26. Income from operations plus non-operating items then equals income before tax exactly, and the effective tax rate was 18.07%.

Second, non-operating items were NT\$95.83bn in 2Q26, or 7.5% of revenue, against NT\$28.83bn and 2.5% in 1Q26. TSMC sold 8.1% of Vanguard International Semiconductor on 15 May 2026. Both of those are disclosed facts. That the disposal contributed to the increase is an inference — TSMC has not disclosed a gain amount in the materials reviewed — and non-operating items also include interest income,

foreign-exchange movements and equity-method income, which are noisy quarter to quarter.

Applying the 1Q26 non-operating ratio of 2.5% to 2Q26 revenue produces a counterfactual EPS of approximately NT\$25.24 against the reported NT\$27.25, implying that roughly 7.4% of the quarter's earnings did not recur from the prior quarter's run-rate. The normalisation is defensible on its own terms: 1Q26's NT\$28.83bn against a cash balance near NT\$3,384bn implies roughly a 3.4% annualised return, consistent with interest income, while the 2Q26 level implies about 10.9%.

This is a single quarter and it is not extrapolated. It is carried forward as a monitoring line rather than a structural conclusion: non-operating items sustained above 3% of revenue would warrant scrutiny of reported earnings growth.

7. The Technical Moat: What We Can and Cannot Identify

TSMC has delivered four consecutive leading-edge node transitions on schedule and was first foundry to volume production at each: N7 in 2018, N5 in Q2 2020, N3 in H2 2022, N2 in 2025. In Q2 2026, 2nm accounted for 3% of wafer revenue, 3nm for 30%, 5nm for 33% and 7nm for 11%, with advanced technologies at 77% in aggregate. Node-ramp dilution has been guided at 3–4 points at each introduction and decayed over roughly 12–14 quarters at N3.

Those are observations. What produces them is not identifiable from public evidence, and the reason is structural rather than a gap that further research would close.

Yield, defect density and cost per good die are the dependent variables of every capability hypothesis, and no firm discloses them at any node. Beyond that, the direction of causation between volume and capability cannot be established: customer commitment at each node precedes same-node capability and follows prior-node capability, so both "volume produces capability" and "capability attracts volume" predict the identical observed sequence at all four transitions. There is no failed TSMC node against which the alternatives could be separated.

Of ten candidate mechanisms examined, one has been substantially weakened and six are untestable rather than merely untested.

Mechanism	Status
First adoption / tool access	Substantially weakened
Supplier co-integration	Plausible, unverified
Yield-learning accumulation	Unresolved
Ecosystem learning	Unresolved
Process integration	Unknown
Defect density and process control	Unknown
Process-window control	Unknown
Engineering staffing density	Unknown
Design-technology co-optimisation	Unknown
Fab-to-fab replication	Unknown

The first row is the one identification result this work produced. The first adopter of the current lithography generation holds approximately 0.4% of the foundry market, while TSMC has deliberately deferred adoption on cost grounds, and tool vendors sell to every qualified buyer. That is an inverse association across a small number of firms, not a controlled test. It establishes that early tool adoption is not sufficient for commercial position. It does not establish that equipment relationships are irrelevant.

It is worth stating plainly what this means for the investment case. High utilisation does not become a moat without an intermediate argument, and the evidence does not support one. Strong margins do not demonstrate superior process capability; they demonstrate that output rose against a largely fixed asset base. The residual capability advantage is real — it is what customers pay for — and no outside party can size it.

8. EUV and Process Technology

The single largest obstacle to identifying the capability mechanism is lithography generation, and it operates in the direction that would flatter a learning-curve explanation.

KEY TAKEAWAY

A horizontal timeline showing the progression of semiconductor technology nodes. The timeline is a solid black line with four black dots representing key nodes. Above the dots are the node names: N7, N5, N3, and N2. Below the dots are the corresponding years: 2018, Q2 2020, H2 2022, and 2025. A horizontal bracket spans from the N3 node to the N2 node, with the text 'N3 dilution decayed over ~12–14 quarters' written below it. Below the timeline, there is a section titled 'LITHOGRAPHY' in bold. Under this title, there is a horizontal bar divided into two sections. The left section is white and labeled 'DUV multi-patterning'. The right section is light gray and labeled 'EUV (introduced at N7+; used from the start at N5 onward)'. At the bottom of the slide, there is a dashed horizontal line, and below it, a light gray box with diagonal hatching containing the text 'Capability mechanism producing these transitions: not identified from public evidence'.

N7 **N5** **N3** **N2**

2018 Q2 2020 H2 2022 2025

N3 dilution decayed over ~12–14 quarters

LITHOGRAPHY

DUV multi-patterning EUV (introduced at N7+; used from the start at N5 onward)

Capability mechanism producing these transitions: not identified from public evidence

Source: node-timeline and dilution disclosures, §7–8. Observed dates and guided dilution decay only; no causal mechanism is implied or scored.

The same confounder contaminates the competitor evidence. Intel confirmed 10nm difficulties in July 2015 and attributed them to multi-patterning defect density. Both Intel's 10nm and GlobalFoundries' 14nm failures sit in the pre-EUV window. They are not independent observations and may constitute a single industry-wide technological shock rather than two firm-specific failures. Treating them as two overstates the evidence.

9. Packaging and Ecosystem

This matters for the wafer arithmetic. Dividing total revenue by wafer shipments produces a figure that rises when packaging grows faster than wafers, with no change in wafer pricing or mix. Stripping packaging and other services out, revenue per wafer grew approximately 8.6% in H1 2026 against the blended 11.2%. The correction is stable: across packaging-and-other assumptions of 11%/13%, 13%/15% and 15%/17%, the growth figure comes out at 8.7%, 8.6% and 8.6% respectively.

10. Competitive Threats

Intel Foundry is best measured against itself, which avoids the comparability problems of a cross-company revenue comparison. FY2025 segment revenue was US\$17.826bn, implying a quarterly average of US\$4.457bn. Q2 2026 external-customer revenue was US\$293m, or 6.6% of its own segment. H1 2026 external revenue was US\$467m, implying US\$174m in Q1 and roughly 68% quarter-on-quarter growth from a small base. The FY2025 segment operating loss was US\$10.318bn. No comparison to TSMC's total revenue is drawn, because that figure includes packaging, mask, test and mature nodes, and no clean comparable metric exists at the leading-edge level.

Intel serves as a warning case rather than a competitive read. It held the largest captive leading-edge volume in the industry at 14nm and failed at 10nm for roughly four years. That establishes that volume with committed customers is not sufficient for capability. It does not establish what is.

Samsung Foundry held Apple and Qualcomm at 14nm and lost both around 2016–17, then reached gate-all-around at 3nm ahead of TSMC without customers following, amid reported leakage problems. The episode is heavily confounded: being first to a transistor architecture is not being better, so it does not test whether capability attracts customers.

GlobalFoundries provides the one mechanism test that succeeds. It halted 7nm development in August 2018 with AMD committed, stating it "did not plan 7nm for high volume," having already seen its own 14nm development fail. Insufficient aggregation makes leading-edge reinvestment uneconomic. That is a claim about funding rather than learning, and it is the only version of the scale argument with supporting evidence.

On neutrality: TSMC has not entered a customer's product market or taken equity in a customer in 39 years. But Samsung aggregated Apple and Qualcomm while integrated, so neutrality is not strictly necessary for aggregation — at most for aggregation at TSMC's scale, which is a materially weaker claim. Neutrality is a permissive condition, not a demonstrated cause. It would be falsified if an integrated competitor operating a firewalled foundry reached double-digit external revenue as a share of its own segment while retaining its product business, and Intel's structure is roughly two years old against a two-to-four-year design-win lag, which is too early to judge.

A useful discipline is to ask which parts of this argument survive substituting another advanced foundry for TSMC.

Argument	Survives substitution?
Fixed-cost absorption from high utilisation	No
Capital rule producing opposite actions	No
Four consecutive on-schedule node transitions	No
Switching cost during a design port	Partially
Convexity of yield economics at large die	Yes (industry physics)
Neutrality as a permissive condition	Only for a demerged foundry
The unidentified capability mechanism	Yes (shared unknown)

Four of seven break, two are industry physics rather than firm-specific, and one is a shared unknown. The thesis is TSMC-specific at the points that matter, and the concession is that the yield-economics and incumbency arguments are generic.

11. Key Thesis Breakers

The following are ranked by observability and time to detection rather than by probability. No probability is assigned to irreducible geopolitical risk.

Breaker	Detection	Current reading
Revenue growth below depreciation compound	1–2 quarters	H1 2026 +35.6%
Sequential revenue per wafer stalls	1 quarter	Q1→Q2 +7.8%
Shipments stall with capex sustained	1–2 quarters	H1 2026 +22.0%
Aggregation counts fall as revenue rises	12 months	All three rising
Capital intensity outside 30–38%	2 quarters	35.2% in H1 2026
Node dilution fails to decay	3–4 years	N3 decayed on precedent
Neutrality erosion	Immediate	Intact for 39 years
Overseas dilution exceeds guidance by >1pp	2–4 quarters	Guided 2–3pp widening to 3–4pp
Competitor external revenue toward parity	2–4 years	6.6% of own segment
Structurally higher cost of capital	Not observable	Dominant swing factor
Taiwan discontinuity	Not observable in advance	No probability assigned

Two of these deserve elaboration.

The first breaker is the bear mechanism and it operates without any competitive event. Depreciation compounds on capital already committed. If revenue growth falls below that compound while capital intensity holds, gross margin compresses arithmetically.

The last is deliberately left unparameterised. All operating scenarios in this report assume continuity. In a material-disruption state, depreciation continues on committed assets while output stops, and with depreciation at 13–20% of revenue, a supply interruption destroys margin considerably faster than a demand interruption. The FY2023 downturn, used throughout this work as the cyclical calibration, was demand-driven and does not bound that case. In a severe-disruption state the discounted cash flow framework does not apply at all; value reduces to overseas assets and net cash.

12. 2026–2030 Operating Scenarios

FY2026 is anchored on reported H1 results and Q3 guidance.

	H1 2026	FY2026E
Revenue, NT\$bn	2,404.48	~5,450
Gross margin	66.99%	~66.1%
Operating margin	59.29%	~58.0%
EPS, NT\$	49.33	~107
Wafer shipments, Kpcs	8,510	~17,700
Capital expenditure, NT\$bn	846.76	~1,971
Free cash flow, NT\$bn	635.57	~1,320

Net cash at 2Q26 sits between NT\$2.2tn and NT\$2.7tn. Cash and marketable securities were NT\$3,518.01bn and long-term interest-bearing debt NT\$864.27bn, but the short-term debt component within current liabilities of NT\$1,857.76bn is not separately disclosed. The range is worth 0.7% of market capitalisation and is immaterial to the conclusion.

All three scenarios hold capital intensity at 34% from 2027. There is no capital-expenditure reduction in the bear case, and therefore no artificial free-cash-flow stabiliser.

Scenario	Revenue growth 2027–30	Utilisation path
Bear	+10%, +5%, +5%, +5%	~93% → ~76%
Base	+25%, +25%, +25%, +20%	~90–95% held
Bull	+30%, +30%, +25%, +20%	At ceiling

The bear case deserves explanation, because its cash-cost assumption is not defensible on inflation alone. Absolute cash cost growing at roughly twice revenue only makes sense through utilisation deleverage. Holding capital intensity at 34% while revenue grows 5% builds capacity roughly 9% a year against shipments growing 2–4%, which takes utilisation from approximately 93% to 76% over the period. With a semi-fixed cash-cost base, cash cost per wafer rises even as absolute volume grows. That is the mechanism, and it is the same utilisation event that drives the depreciation ratio — consistent with the argument made throughout this report.

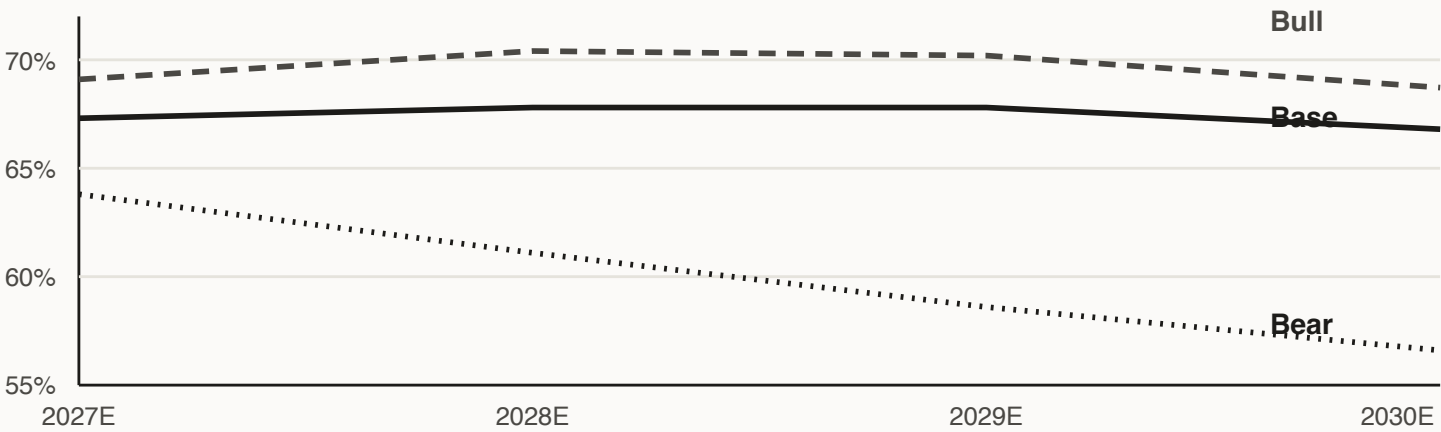
Bear case, NT\$bn	2027E	2028E	2029E	2030E
Revenue	6,029	6,330	6,647	6,979
Depreciation	953	1,084	1,248	1,425
Depreciation % revenue	15.8%	17.1%	18.8%	20.4%
Gross margin	63.8%	61.1%	58.6%	56.6%
EPS, NT\$	110	109	109	110
Free cash flow	1,556	1,555	1,606	1,685
Base case, NT\$bn	2027E	2028E	2029E	2030E
Revenue	6,851	8,563	10,704	12,845
Depreciation % revenue	13.9%	13.1%	13.0%	13.7%
Gross margin	67.3%	67.8%	67.8%	66.8%
EPS, NT\$	134	169	211	249
Free cash flow	1,856	2,291	2,844	3,403
Bull case, NT\$bn	2027E	2028E	2029E	2030E
Revenue	7,125	9,262	11,578	13,893
Gross margin	69.1%	70.4%	70.2%	68.7%
EPS, NT\$	144	192	239	280
Free cash flow	1,862	2,459	3,082	3,676

The bear case is a depreciation and utilisation story rather than a demand collapse. Revenue grows every year. Earnings per share are flat at NT\$109–110 for four consecutive years and gross margin lands at 56.6%, at management's own through-cycle objective of 56% and higher. Four points of gross margin disappear without any competitive event.

EXHIBIT 05 — SCENARIO MECHANISM: GROSS MARGIN, 2027–2030

KEY TAKEAWAY

The bear case is a utilisation and depreciation story, not a demand collapse — revenue grows every year while margin compresses and EPS holds flat at NT\$109–110.



Source: base/bear/bull operating scenarios, §12 and Appendix D. Bear-case EPS is flat at NT\$109–110 across all four years despite revenue growth; see §12 table.

13. Valuation and Market-Implied Requirements

Case	10% / 2.5%	10% / 3%	9% / 3%	8% / 4%
Bear	−65%	−63%	−58%	−41%
Base, 20% CAGR	−44%	−41%	−32%	−3%
Base, 25% CAGR	−38%	−34%	−24%	+11%
Bull, 30% CAGR	−33%	−29%	−18%	+19%

EXHIBIT 04 — WHAT TODAY’S PRICE REQUIRES

KEY TAKEAWAY

Only the 8%/4% discount-rate/terminal-growth combination clears the current price, and only if growth meets or exceeds management’s stated objective.

	10% / 2.5%	10% / 3%	9% / 3%	8% / 4%
Bear	−65%	−63%	−58%	−41%
Base, 20% CAGR	−44%	−41%	−32%	−3%
Base, 25% CAGR	−38%	−34%	−24%	+11%
Bull, 30% CAGR	−33%	−29%	−18%	+19%

Shading is value vs. market: darkest fails most; boxed cells clear the current price.

Source: discounted-cash-flow output vs. US\$2.19tn market capitalisation; see §2 and §13. Figures unchanged from the base valuation table.

Terminal value represents 86–89% of the discounted cash flow output in every cell. This is arithmetic rather than a modelling artefact: with a four-point spread between discount rate and terminal growth, the terminal multiple is 25 times, and no plausible extension of the explicit forecast period materially reduces its share. For that reason no intrinsic value is asserted anywhere in this report. The model is used to identify the market's required conditions.

A multiple cross-check at US\$422 per ADR gives a different-looking picture:

	2026E	2027E	2028E	2029E	2030E
P/E	24.8×	20.0×	15.9×	12.7×	10.8×
FCF yield	1.9%	2.6%	3.3%	4.1%	4.9%

On the base case the shares trade at roughly 11 times 2030 earnings, which is not obviously expensive. On discounted cash flow the price clears only at 8% with 4% terminal growth. The two views reconcile through the discount rate and nowhere else. In the bear case, 2030 EPS of NT\$110 implies 24.6 times 2030 earnings after four years without growth.

The appropriate discount rate cannot be established from this framework. A Taiwan risk-free rate near 1.5%, an equity risk premium near 5% and a beta near 1.1 produce roughly 7%, and any premium for the concentration of leading-edge capability in a single jurisdiction takes it above 8%. How large that premium should be is an investor judgment, not an analytical output, and this report does not manufacture one.

14. Monitoring Dashboard

Indicator	Current	Trigger	Frequency
Sequential revenue per wafer	+7.8% Q1→Q2	Below +4%, two quarters	Quarterly
Wafer shipments	8,510 Kpcs H1	Flat/falling, two quarters, capex sustained	Quarterly
Capital intensity	35.2% H1	Outside 30–38%, two quarters	Quarterly
Gross margin vs. guidance	66.99% H1	Missing by more than 2pp	Quarterly
Non-operating items	7.5% of revenue, 2Q26	Sustained above 3% of revenue	Quarterly
Free cash flow margin	26.4% H1	Below 20% with capex in band	Quarterly
Competitor external revenue	6.6% of own segment	Sustained growth toward parity	Quarterly
Overseas dilution vs. guidance	2–3pp widening to 3–4pp	Exceeding guidance by 1pp	Quarterly
Depreciation ÷ revenue	~15.0% FY2026E	Rising two consecutive years	Annual
Customer/product/technology counts	534 / 12,682 / 305	Any falling as revenue rises	Annual
Total capacity	Above 17m, FY2025	Flat two years, capex sustained	Annual
Fab table by node	All ≤3nm in Taiwan	A ≤2nm node outside Taiwan	Annual
Advanced-node mix	77% of wafer revenue	Qualitative; no threshold identifiable	Quarterly
Packaging capacity	"Limited by customers' growth"	Qualitative; capacity not disclosed	Quarterly
Neutrality	Intact, 39 years	Product entry or customer equity	Event

Four variables remain unresolved and are tracked rather than estimated.

Variable	Resolving disclosure	Interpretation
Quarterly 2026 depreciation	Full quarterly statements; annual functional-expense note	Back-loaded phasing implies the FY2026 expansion is depreciation-led and reverses on schedule; front-loaded implies cash-cost leverage that reverses on growth
Process-capability mechanism	None available	No threshold can be set; qualitative monitor only
Capex per unit of leading-edge capacity	None available	Intensity above 38% while revenue per wafer stalls would be adverse
Discount rate and geopolitical risk	Fab table by node	A ≤2nm node outside Taiwan would reduce concentration and support a lower rate

EXHIBIT 06 — THESIS-BREAKER MAP BY CATEGORY		
KEY TAKEAWAY		
Every breaker below is already tracked in §11 and §14; grouping by category shows how much of the thesis rests on observable quarterly data versus what cannot be observed at all.		
QUARTERLY OPERATING INDICATORS		
Revenue growth below depreciation compound		1–2 quarters
Sequential revenue per wafer stalls		1 quarter
Shipments stall with capex sustained		1–2 quarters
CAPITAL-ALLOCATION INDICATORS		
Capital intensity outside 30–38%		2 quarters
Overseas dilution exceeds guidance by >1pp		2–4 quarters
COMPETITIVE INDICATORS		
Competitor external revenue toward parity		2–4 years
STRUCTURAL INDICATORS		
Aggregation counts fall as revenue rises		12 months
Node dilution fails to decay		3–4 years
Neutrality erosion		Immediate
UNOBSERVABLE / NON-PARAMETERISED		
Structurally higher cost of capital		Not observable
Taiwan discontinuity		No probability assigned
Source: §11 (Key Thesis Breakers) and §14 (Monitoring Dashboard), regrouped by category. No new indicators, thresholds or probabilities are introduced.		

15. Investment Committee Decision Card

Recommendation. The available public evidence cannot establish an actionable edge at US\$422.

Thesis. TSMC's observable economic engine — aggregation to utilisation to fixed-cost absorption to surplus to reinvestment — is sound and largely arithmetic. The technical mechanism beneath it is not identifiable from public evidence: one of ten candidate sub-mechanisms has been substantially weakened and six are untestable. At today's price the base case clears at 8% discount with 4% terminal growth and

fails at 9% with 3%, with terminal value at 86–89% of output. The decision turns on a discount rate this framework has no defensible basis to set.

Supporting evidence. H1 2026 shipments rose 22.0% with blended revenue per wafer up 11.2%, and the Q1-to-Q2 sequential pattern reconciles FY2026 guidance without a step-change. Capital intensity of 35.2% sits inside a rule that produced opposite actions on opposite signals in 2023 and 2026. The aggregation base expanded on all three disclosed measures.

Counter-evidence. Utilisation is at or near its ceiling and the next capacity tranche is guided to 2028–29. In the bear case, utilisation falls to roughly 76%, depreciation reaches 20.4% of revenue and EPS is flat for four years. Approximately 7.4% of 2Q26 earnings did not recur from the prior quarter's run-rate.

Market-implied requirements. At 9%/3%, a revenue CAGR near 35% through 2030. At 8%/4%, management's 25% objective, met. At 20% growth, even 8%/4% leaves the shares roughly fair. Capital intensity must hold in the mid-30s despite disclosed rising unit capital cost. All scenarios assume operational continuity in Taiwan.

Unresolved variables. Quarterly 2026 depreciation, which swings the H1 margin decomposition between 59/41 and 30/70. The process-capability mechanism. Capital expenditure per unit of leading-edge capacity. The appropriate discount rate.

Upgrade conditions. A 2nm-or-below node entering volume production outside Taiwan. Two further quarters at the current sequential run-rate. Capital intensity holding mid-30s through a normalised-growth year. Consensus estimate data enabling a differentiation test.

Downgrade conditions. Shipment growth stalling with capital intensity sustained. Depreciation as a share of revenue rising two consecutive years. Overseas dilution exceeding guidance by more than one point. Competitor external foundry revenue growing sustainably toward segment parity.

Next observable evidence. Q3 2026 results, covering revenue against guided US\$44.6–45.8bn, gross margin against 65–67%, shipments, capital intensity and non-operating items. The FY2026 annual filing, covering capacity, full-year shipments, aggregation counts, the fab table by node and the functional-expense note. Full quarterly financial statements, which contain the depreciation line. Quarterly competitor external foundry revenue.

Dimension	Confidence
Accounting	High
Economic engine	High
Technical moat	Low
Competitive position	Medium
Forecast	Medium
Valuation	Low
Geopolitical risk	Not parameterised

Appendices

Appendix A. Methodology and Formulas

Revenue decomposition. $\text{Revenue} \approx \text{Wafer Shipments} \times \text{Revenue per Wafer}$.

Growth bridge. $(1 + \text{revenue growth}) \approx (1 + \text{shipment growth}) \times (1 + \text{revenue-per-wafer growth})$. Verified on H1 2026 year-on-year ($1.220 \times 1.112 = 1.356$ against 1.356 reported) and on Q1-to-Q2 2026 ($1.039 \times 1.078 = 1.120$ against 1.120 reported).

Volume share of growth. $\text{Volume share} = \ln(1 + \text{shipment growth}) \div \ln(1 + \text{revenue growth})$. The log-ratio form is used because the two effects are multiplicative; an arithmetic split would not sum to unity. This produces 65% for the year-on-year comparison and 34% for the sequential.

Ex-packaging revenue per wafer. $(\text{Total revenue} \times (1 - \text{packaging and other share})) \div \text{wafer shipments}$. Packaging share is disclosed verbally in annual materials; mask, test and other are not disclosed and are assumed at 3%.

Discounted cash flow. $\text{Enterprise value} = \sum \text{FCF}_t \div (1 + \text{WACC})^t + \text{TV} \div (1 + \text{WACC})^n$, where $\text{TV} = \text{FCF}_{(n+1)} \div (\text{WACC} - g)$.

The terminal-value share is structurally high in this application. With $\text{WACC} - g$ of four points, the terminal multiple is 25 times terminal free cash flow, against a present value of four explicit years amounting to roughly three times average annual free cash flow. Extending the explicit period does not materially reduce the share, because the additional years are themselves projections rather than observations. This is why the model is presented as a requirement engine rather than a valuation.

Appendix B. Valuation Mechanics

Market capitalisation of approximately US\$2.19tn is converted at NT\$32 per US dollar for the valuation comparison. Reported average rates were 31.59 in 1Q26 and 31.60 in 2Q26; the FY2026 model uses 31.8.

Net cash is taken within a range of NT\$2.2–2.7tn. The upper bound deducts only long-term interest-bearing debt of NT\$864.27bn from cash and marketable securities of NT\$3,518.01bn. The lower bound assumes a portion of current liabilities of NT\$1,857.76bn is interest-bearing; the split is not disclosed. The difference is 0.7% of market capitalisation.

The required-conditions table in Section 2 holds the base-case cost structure, capital intensity and terminal assumptions constant and solves only for the revenue growth rate that equates the discounted cash flow to the current market capitalisation.

Appendix C. Depreciation Model

The schedule applies a machinery share to a rolling five-year capital-expenditure average, plus a building component: $\text{D\&A}(t) = 0.68 \times \text{mean}(\text{capex from } t-1 \text{ to } t-5) + 105$, in NT\$bn.

Historical capital-expenditure inputs, NT\$bn: 2021 approximately 840; 2022 1,083; 2023 950; 2024 956; 2025 1,272. Pre-2021 inputs used in back-testing are converted from US dollar figures at assumed rates and are the source of the largest back-test errors.

Two limitations govern how much weight this schedule can bear. First, with four free parameters and five historical observations, twenty of 420 parameter combinations achieve better than 6% mean absolute error, so the fit is not discriminating. Second, a genuine out-of-sample test — calibrating on 2021, 2022 and 2024, then predicting 2025 without reference to the outcome — produces an 8.4% error. Sensitivity to the assumed useful life is material: a one-year change moves mean error from 4.2% to 6.3%, and a one-year change in the in-service lag moves it to 9.8%.

Appendix D. Scenario Assumptions

Common to all scenarios: capital intensity of 34% from 2027 (36% in FY2026, as guided); tax at 18%; non-operating income at 2.5% of revenue, matching the 1Q26 run-rate; 25,932m shares held flat with no buyback; working-capital drag of 5.4% of pre-working-capital operating cash flow, calibrated to FY2025 where reported operating cash flow of NT\$2,275.00bn ran below net income plus depreciation of NT\$2,405.98bn.

Scenario	Cash cost of revenue	Operating expenses
Bear	21.5% rising to 24.5%	8.5% rising to 9.0%
Base	19.8% rising to 20.5%	8.0%
Bull	18.5% held	7.8% falling to 7.5%

Capacity and utilisation are not forecast beyond FY2026. Disclosed capacity is rounded to the nearest million 12-inch equivalents, producing an eleven-point band in implied FY2026 capacity growth; projecting that figure five years forward would be false precision. Utilisation enters the model through the cash-cost ratio, which is the observable channel.

Appendix E. Evidence and Source Register

Primary sources. TSMC Form 6-K filings for 1Q26 and 2Q26, comprising earnings releases and investor presentations; TSMC Form 20-F for FY2025 and prior years; consolidated financial statement notes, including the disclosure disaggregating depreciation and amortisation between operating costs and operating expenses; quarterly management reports; earnings-call transcripts hosted on TSMC's investor relations site; TSMC technology pages and stated long-term financial objectives.

Competitor primary sources. Intel segment reporting for Foundry revenue, external revenue and segment operating result.

Secondary sources. Contemporaneous trade coverage of node timelines, the GlobalFoundries 7nm exit, Intel's 10nm difficulties and Samsung's 3nm programme. Each is dated and is used only where no primary disclosure exists.

Market data. Market capitalisation and ADR price, from third-party providers whose figures span US\$1.90–2.19tn. The upper end of the range is used and the divergence is disclosed.

Derived figures. Utilisation, revenue per wafer, cost per wafer, the margin bridge, capital intensity, the depreciation schedule, all scenario outputs and all valuation outputs are calculated from primary inputs. Every such figure is identified as calculated in the text or table in which it appears.

Appendix F. Calculation Notes and Reconciliations

2Q26 income statement. Revenue NT\$1,270.381bn $\times$ gross margin 67.72% = gross profit NT\$860.311bn, matching the reported figure. Gross profit less operating expenses of NT\$98.98bn gives NT\$761.33bn against reported income from operations of NT\$766.60bn; the NT\$5.27bn difference is other operating income and expenses. Income from operations plus non-operating items of NT\$95.83bn equals income before tax of NT\$862.430bn. Net income of NT\$706.56bn implies an effective tax rate of 18.07%.

FY2025 cash flow. Net income of NT\$1,717.88bn plus depreciation and amortisation of NT\$688.10bn gives NT\$2,405.98bn against reported operating cash flow of NT\$2,275.00bn, a working-capital and other drag of 5.4%. Less capital expenditure of NT\$1,272.41bn gives free cash flow of NT\$1,002.59bn against the reported NT\$1,002.57bn.

FY2026 run-rate reconciliation. Applying Q1-to-Q2 growth rates of 12.04% for revenue and 3.88% for shipments to Q2 2026 gives Q3 revenue of NT\$1,423bn against a guidance midpoint of NT\$1,437bn, and a full-year figure of NT\$5,423bn on 17,693 Kpcs of shipments.

Margin decomposition, FY2024 to FY2025. Two methods are reported and neither is preferred. The per-wafer method holds depreciation per wafer at its FY2024 level and gives 54–61%. The ratio method decomposes gross margin into depreciation-to-revenue and cash-cost-to-revenue and gives 119%, with the cash-cost line a net drag. The divergence arises because a product of two terms does not decompose without an ordering convention.

Appendix G. Definitions and Evidence Grades

Grade	Meaning
A	Disclosed or directly observed in a primary source
B	Calculated from grade A inputs, with the formula shown
C	Strong inference from observed evidence
D	Plausible but unverified
E	Unknown or unidentifiable from public evidence
Guided	Management forward statement; an assertion, not an outturn
Assumed	Explicit modelling choice with no disclosure behind it

Grades do not migrate upward anywhere in this report. Where a claim rests on inference, the inference is stated rather than absorbed into the surrounding prose.

Wafer shipments are 12-inch-equivalent, as disclosed by TSMC, and include conversions from other wafer sizes.

Blended revenue per wafer is total consolidated revenue divided by wafer shipments and is therefore contaminated by advanced packaging, mask, test and other services. **Ex-packaging revenue per wafer** strips those components using disclosed packaging shares and an assumption for other services.

Cash cost of revenue is cost of revenue less depreciation charged to cost of revenue. It is not a signal in its own right: gross margin equals one minus the depreciation ratio minus the cash-cost ratio, so the three are related by identity rather than by independent information.

Capital intensity is capital expenditure divided by revenue, as management defines it.

Eleven claims were withdrawn during the preparation of this research. One forward prediction was made and it was wrong. The mechanism producing TSMC's process capability is unidentified from public evidence; the economic system surrounding it is not.